

1 Examination development

Shared examination blueprint: urology nursing syllabus, textbooks & clinical guidelines, case scenarios, item banks
matched item types across arms (single best answer · multiple response · case analysis · total score 100)

MAS-based AI examination

Structured prompt +
curated knowledge inputs

MAS-based drafting, revision
& answer-key checking

Independent factual screening
by 1 nurse educator

Final AI-generated examination

Expert-developed examination

2-3 senior urology
nurse educators

Same blueprint and
content coverage

Consensus drafting and
internal review

Final expert-developed examination

2 Prospective crossover administration

Participants

- 41 registered urology specialty nurses
- Clinical experience ≥ 6 months
- Written informed consent
- Randomized 1:1 to two sequences

Randomized two-sequence allocation

Sequence A · AI-generated exam (n = 21)

Sequence B · Expert-developed exam (n = 20)

Standardized administration

- Standardized classroom setting
- Closed-book administration
- 90 minutes per examination

3 Scoring, psychometrics & statistical analysis

Scoring

- Objective items: automated
- Case items: 2 blinded nurse raters
- Third-rater adjudication for disagreements

Question quality

- Total score & score by item type
- Difficulty (P) & discrimination (D)
- Internal consistency (Cronbach α)
- Correlation between the two exams

Analysis & outputs

- Difference in total score
- Completion time & inter-rater agreement
- Group comparison / crossover
- Feasibility & equivalence of AI